



ACTAI • WEEK 03 • LECTURE 02

Apps That Think

Putting real AI inside the apps you build — with a free Gemini API key.

A live demo lecture — build along in AI Studio

WHERE WE ARE

Three ideas, now combined



Last week

You built apps that DO things —
timers, lists, calculators

+



Yesterday

You instructed an AI with a Gem —
a system prompt

+



Today

Combine them: apps that THINK,
with AI built in

THE CORE IDEA

What 'AI inside an app' means



Your app is the messenger. The thing that lets it talk to Gemini is a free API key.

REMEMBER THIS FROM YESTERDAY?

The system prompt is the app's brain



The instructions you wrote for a Gem are the EXACT same thing you give the AI inside your app. You already know how to do this.

- Set the role: 'You are a helpful recipe assistant'
- Set the rules: 'Only suggest recipes from these ingredients'
- Set the format: 'Reply with a title, steps, and a shopping list'
- Change the prompt → change what the whole app does

DEMO LIVE: Show a Gem's instructions next to an app's prompt — they're the same.

THE MISSING PIECE

What is an API key?



An API key is like a password that lets YOUR app talk to Gemini.



It's free

Google gives everyone a free Gemini key — no credit card.



It's quick

About five minutes to get one and paste it in.



Keep it private

Treat it like a password — don't post it publicly.

DO THIS ONCE

Get your free Gemini API key

1. Go to aistudio.google.com/apikey
2. Sign in with any Google account
3. Click **Create API key**
4. Let it make a new project (automatic)
5. Copy the key (it starts with **AIza...**)
6. Paste it into your app when it asks



No credit card

Just a Google account. Everyone in this room can get their own, free.



Treat it like a password

Don't share it or post it online. If it leaks, delete it on the same page and make a new one.

What 'free' gets you



The free tier

- Permanent — no trial, no expiry
- Runs on Gemini Flash (fast & capable)
- A few requests/min, hundreds per day
- Plenty for learning & this whole course



Keep in mind

- Each person has their own key & quota
- Heavy bursts can hit a rate limit — wait & retry
- On free tier, don't paste anything sensitive
- Top models (Pro) need a paid plan — not needed here

ALTERNATIVES

Other places to get an API key

FREE · NO CREDIT CARD (LIKE GEMINI)



Groq

groq.com

Blazing fast; runs open models like Llama.



OpenRouter

openrouter.ai

One key → 300+ models in one place.



Mistral

mistral.ai

European & GDPR-friendly; free tier.

PREMIUM · PAY-AS-YOU-GO (TOP QUALITY, NO FREE TIER)



OpenAI

platform.openai.com

GPT models. Top quality, pay per use — no free tier.



Anthropic

console.anthropic.com

Claude models. Great writing & reasoning. Paid.

All work the same way: sign up → get a key → paste it in. Gemini's free tier is just the easiest start.

Choosing a model in your request

You pick the model by its NAME in the request. Swap the name — same key — to switch models.

IN YOUR REQUEST

model: "gemini-2.5-flash" ← change this to switch models

gemini-2.5-flash	Fast & capable — your free default	FREE
gemini-2.5-flash-lite	Even faster, for lighter tasks	FREE
gemini-3.1-pro	Smartest — deep reasoning	PAID

Rule of thumb: start with Flash. Reach for Pro only when a task needs deeper reasoning — faster & cheaper usually wins. Other providers work the same way (gpt-..., claude-...).

How to build an AI app

1. Open AI Studio's build area (aistudio.google.com)
2. Describe your app AND its AI feature in one prompt
3. When it asks, paste your free API key
4. Try the app in the live preview
5. Refine by talking — tweak the AI's instructions
6. Share or deploy when you're happy

WATCH THIS

Live build: an AI Trip Planner

Paste this into AI Studio, add your key, and watch it build an app that plans trips with AI.

BUILD PROMPT — paste into AI Studio

An AI trip planner. The user enters a destination, number of days, and their interests (food, history, nature, nightlife). Use Gemini to generate a day-by-day itinerary with morning, afternoon and evening activities, plus one local food tip per day. Clean, friendly layout with a 'Plan my trip' button.

THEN TRY IT

Destination: Tokyo • Days: 3 • Interests: food + culture

- Watch the AI write a full 3-day itinerary, live, inside your app.

Change its brain, change the app



The app's behaviour lives in its prompt. Edit the instructions and the whole app changes personality — no new code.

- “Make the itinerary budget-friendly”
- “Add estimated costs for each day”
- “Write it in a fun, casual tone”
- “Reply in the user's chosen language”

DEMO LIVE: Tell the trip planner to add a daily budget, and re-run it.

BUILD THESE LIVE

AI app ideas – set 1



Fridge Chef

BUILD PROMPT: An AI recipe app. The user types the ingredients they have. Use Gemini to suggest one recipe with a title, steps, and a short shopping list for anything missing.



Study Quiz Coach

BUILD PROMPT: An AI study coach. The user enters any topic. Use Gemini to make a 5-question multiple-choice quiz, let them answer, then grade and explain each one.



Cover Letter Tailor

BUILD PROMPT: The user pastes a job description and a few lines about themselves. Use Gemini to write a tailored one-page cover letter plus 3 interview questions to prepare.

BUILD THESE LIVE

AI app ideas – set 2



Choose-Your-Adventure

BUILD PROMPT: An AI story game. The user picks a genre and a character. Use Gemini to tell an interactive story, pausing each step to offer 2-3 choices that shape what happens next.



Chat With My Notes

BUILD PROMPT: The user pastes an article or their notes, then asks questions. Use Gemini to answer ONLY from the pasted text, and say so if the answer isn't there.



Language Tutor

BUILD PROMPT: The user picks a language and a situation (e.g. ordering food). Use Gemini to give 8 useful phrases with pronunciation tips, then run a short practice dialogue.

THE REAL WOW

Apps that can SEE

Gemini can read images too — so your app can take a photo as input. Same idea, more magic.



Snap-a-Recipe

BUILD PROMPT: The user uploads a photo of their fridge. Use Gemini to identify the ingredients it sees and suggest a recipe they can make right now.



Homework Helper

BUILD PROMPT: The user uploads a photo of a maths or science problem. Use Gemini to explain how to solve it step by step – without just giving the final answer.

KEEP IT REAL

Limits, cost & safety



Free is for learning & prototypes

Generous, but big public apps need a paid plan eventually.



Rate limits exist

Rapid-fire requests can throttle — wait a second and retry.



Always sanity-check AI output

It can be wrong or make things up. Verify what matters.



Protect your key & data

Never post your key; don't feed sensitive info on free tier.



LECTURE RECAP

Your apps can think now.

A free API key + a system prompt = an app powered by real AI. You went from prompt, to Gem, to a thinking app.

Homework · get your key and build one AI app. **Tomorrow** — going further with AI apps.